

LAPORAN PRAKTIKUM 3

Batch Data Analytics & Visualization Layer

Untuk memenuhi tugas mata kuliah Teknologi Big Data

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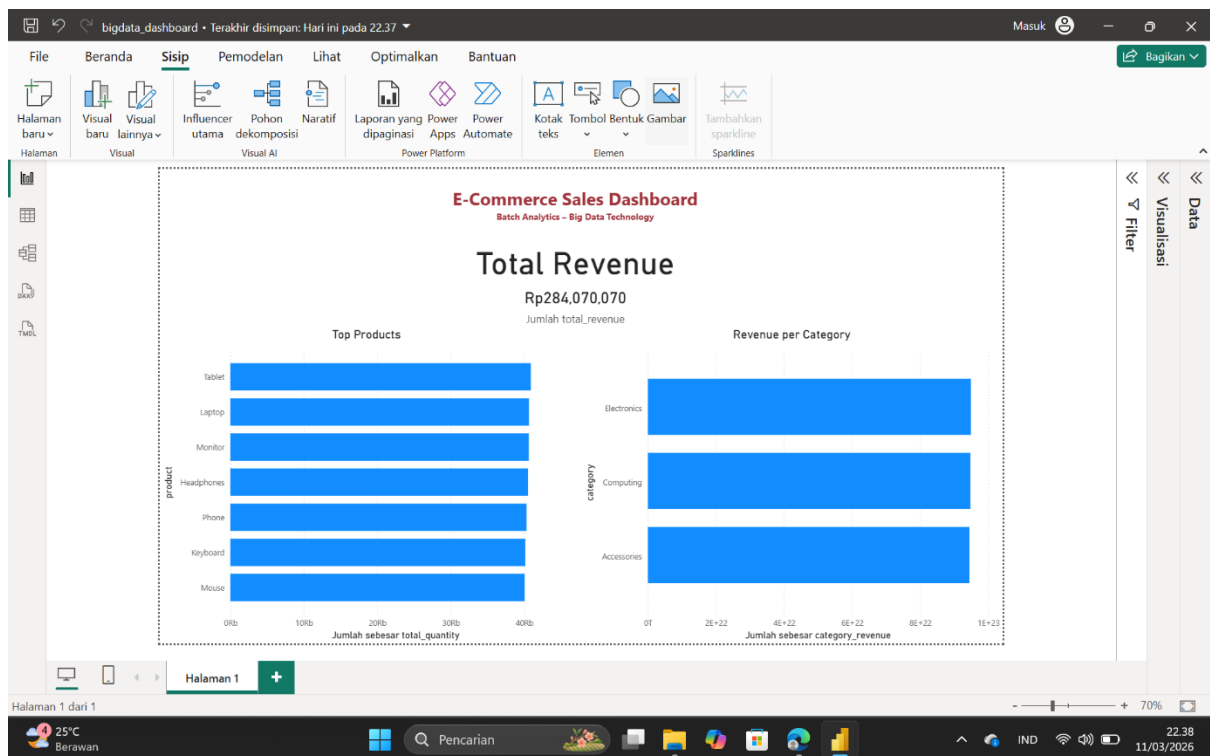
Oleh:

Siti Magfiratun Warahmah

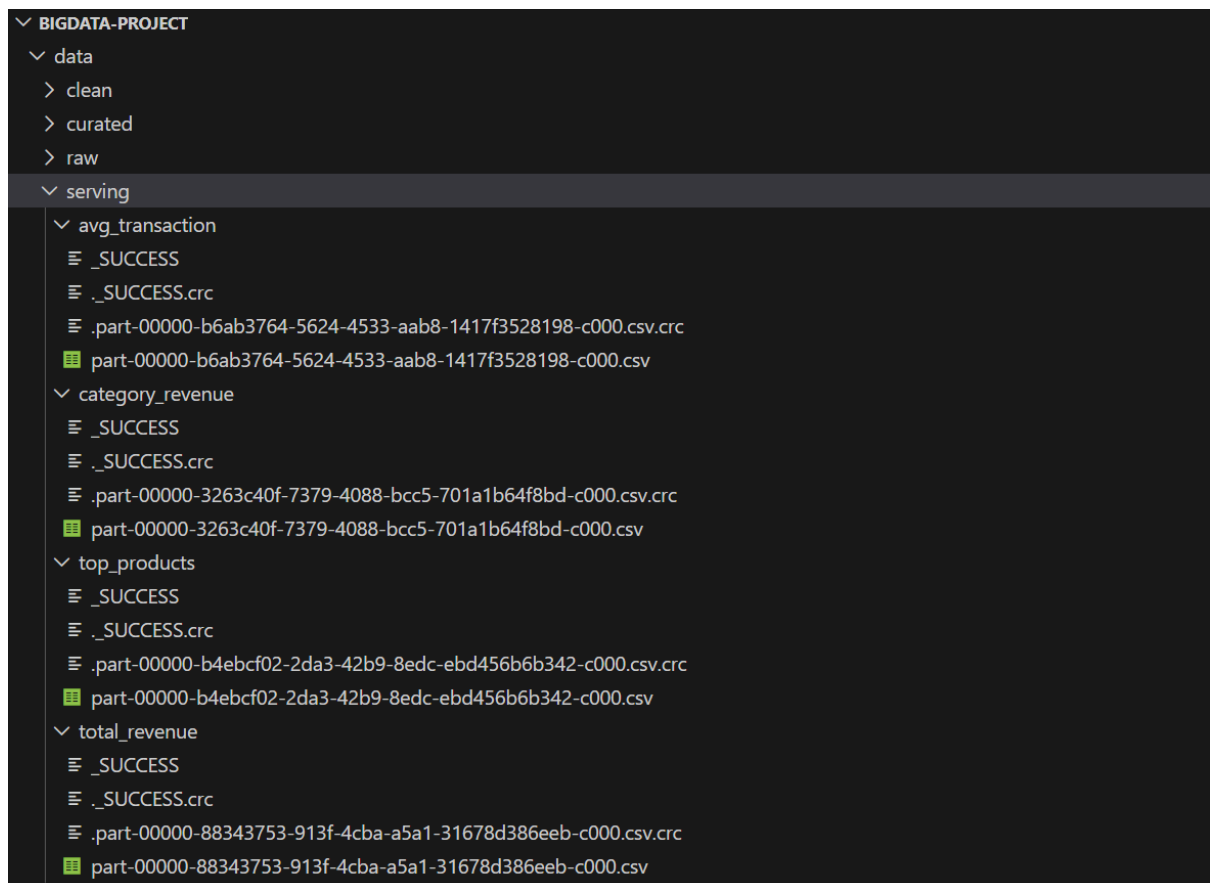
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**PROGRAM STUDI TEKNOLOGI INFORMASI
FAKULTAS DAKWAH DAN ILMU KOMUNIKASI
UNIVERSITAS ISLAM NEGERI ANTASARI BANJARMASIN
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Screenshot Screenshot dashboard Power BI

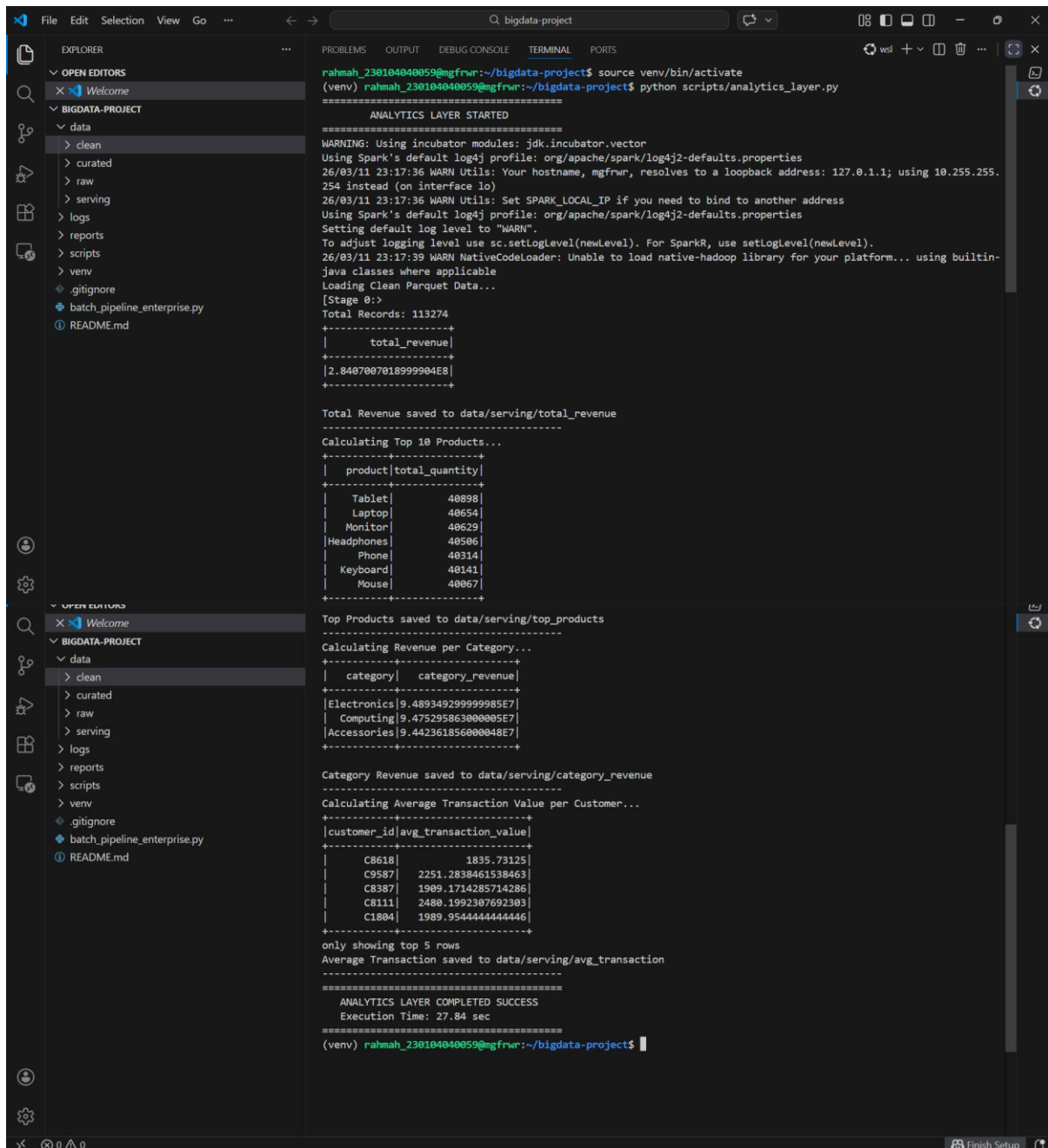


Screenshot folder serving dataset: data/serving



Screenshot Screenshot terminal saat menjalankan:

python scripts/analytics_layer.py



```
File Edit Selection View Go ... bigdata-project
EXPLORER
  OPEN EDITORS
    Welcome
  BIGDATA-PROJECT
    data
      clean
      curated
      raw
      serving
      logs
      reports
      scripts
      venv
    .gitignore
    batch_pipeline_enterprise.py
    README.md
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
rahmah_230104040059@mgfrwr:~/bigdata-project$ source venv/bin/activate
(venv) rahmah_230104040059@mgfrwr:~/bigdata-project$ python scripts/analytics_layer.py

ANALYTICS LAYER STARTED
=====
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
26/03/11 23:17:36 WARN Utils: Your hostname, mgfrwr, resolves to a loopback address: 127.0.0.1; using 10.255.255.254 instead (on interface lo)
26/03/11 23:17:36 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/03/11 23:17:39 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Loading Clean Parquet Data...
[Stage 0:]>
Total Records: 113274
+-----+
| total_revenue |
+-----+
| 2.8407007018999904E8 |
+-----+

Total Revenue saved to data/serving/total_revenue

Calculating Top 10 Products...
+-----+
| product | total_quantity |
+-----+
| Tablet | 40898 |
| Laptop | 40654 |
| Monitor | 40629 |
| Headphones | 40506 |
| Phone | 40314 |
| Keyboard | 40141 |
| Mouse | 40067 |
+-----+

Top Products saved to data/serving/top_products

Calculating Revenue per Category...
+-----+
| category | category_revenue |
+-----+
| Electronics | 9.489349299999985E7 |
| Computing | 9.475295863000005E7 |
| Accessories | 9.44236185600048E7 |
+-----+

Category Revenue saved to data/serving/category_revenue

Calculating Average Transaction Value per Customer...
+-----+
| customer_id | avg_transaction_value |
+-----+
| C8618 | 1835.73125 |
| C9587 | 2251.2830461538463 |
| C8387 | 1909.1714285714286 |
| C8111 | 2480.1992307692303 |
| C1804 | 1989.9544444444446 |
+-----+

only showing top 5 rows
Average Transaction saved to data/serving/avg_transaction

=====
ANALYTICS LAYER COMPLETED SUCCESS
Execution Time: 27.84 sec
=====
(venv) rahmah_230104040059@mgfrwr:~/bigdata-project$
```

Berdasarkan praktikum yang telah dilakukan, dapat disimpulkan bahwa proses analisis data dalam sistem Big Data memerlukan beberapa tahapan pipeline mulai dari pemrosesan data hingga visualisasi informasi. Pada praktikum ini, data yang telah dibersihkan diproses menggunakan PySpark pada tahap analytics layer untuk menghasilkan metrik bisnis seperti total revenue, produk terlaris, dan pendapatan berdasarkan kategori. Hasil analisis tersebut kemudian disimpan pada serving layer dan diintegrasikan dengan Power BI untuk membangun dashboard visual yang informatif. Melalui dashboard ini, data yang sebelumnya kompleks dapat disajikan dalam bentuk visualisasi yang lebih mudah dipahami sehingga membantu pengguna dalam memperoleh insight dan mendukung proses pengambilan keputusan berbasis data.